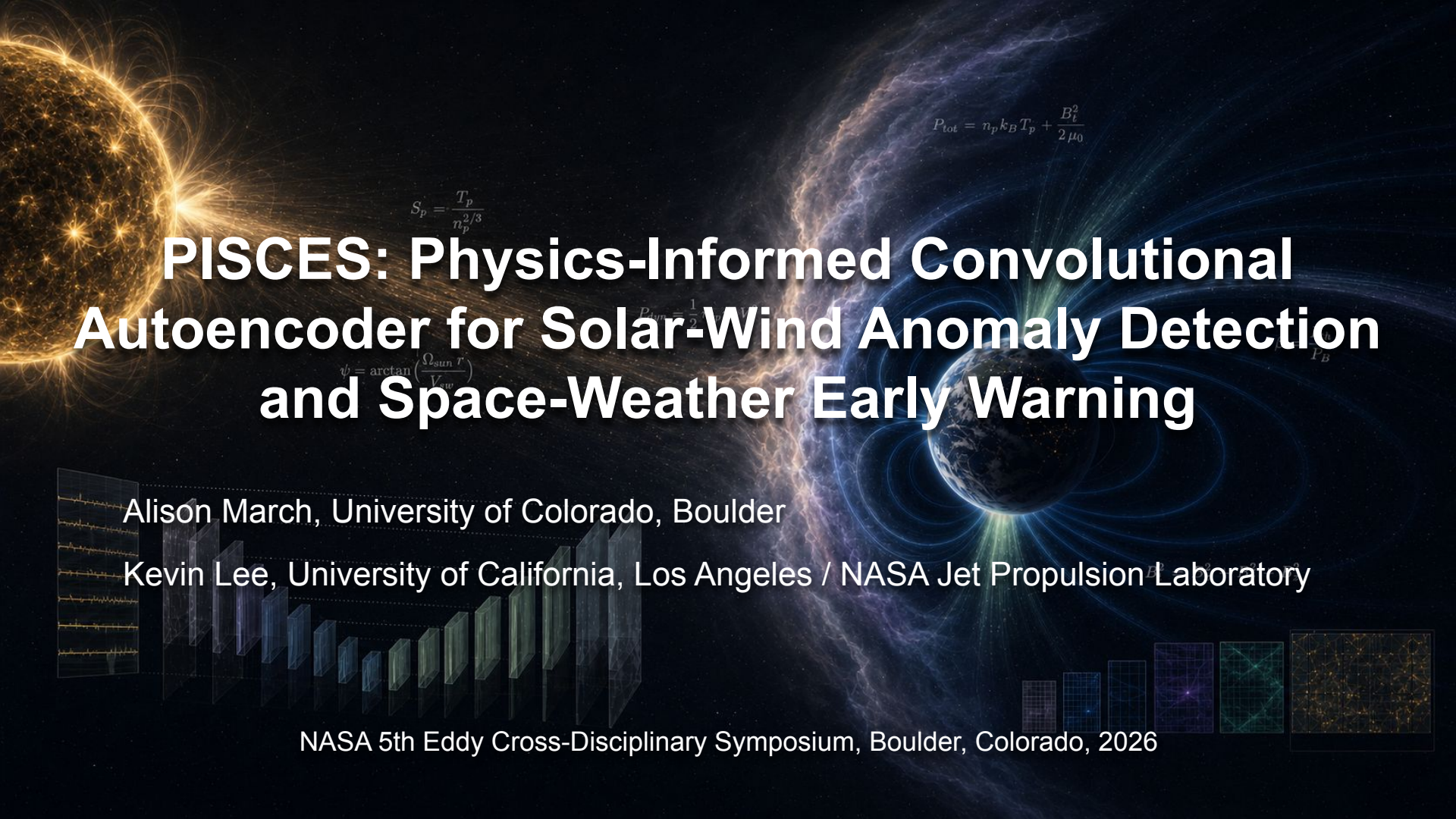




$S_p = \frac{T_p}{n_p^{2/3}}$

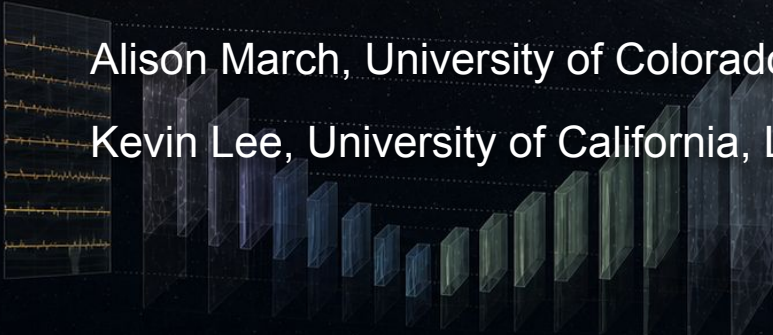
$P_{\text{tot}} = n_p k_B T_p + \frac{B_i^2}{2\mu_0}$

$\psi = \arctan\left(\frac{\Omega_{\text{sun}} r}{V_{\text{sw}}}\right)$

$P_{\text{sun}} = \frac{1}{2}$

PISCES: Physics-Informed Convolutional Autoencoder for Solar-Wind Anomaly Detection and Space-Weather Early Warning

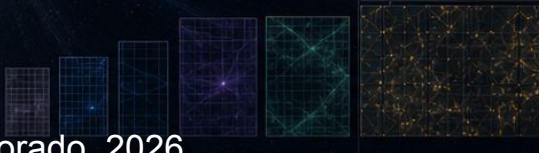
P_B



Alison March, University of Colorado, Boulder

Kevin Lee, University of California, Los Angeles / NASA Jet Propulsion Laboratory

NASA 5th Eddy Cross-Disciplinary Symposium, Boulder, Colorado, 2026



Background and Data

Data and Input Specification

Category	Details
Training Data	NASA OMNI high-resolution solar wind, 2005–2024 (1-min cadence at L1)
Real-time Input	NOAA SWPC RTSW JSON feeds
Evaluation Catalogs	DONKI HSS & IPS; Richardson–Cane near-Earth ICME catalog
Train Split	2005–2015
Validation Split	2016–2017
Test Split	2018–2024
Training Labels	None (unsupervised)
Evaluation Labels	DONKI IPS/HSS and Richardson–Cane ICME

Input Channels (7 Features)

Domain	Channels
Magnetic Field	B_x, B_y, B_z, B_t [nT]
Plasma	n_p [cm ⁻³], V [km s ⁻¹], T_p [K]

Preprocessing Pipeline

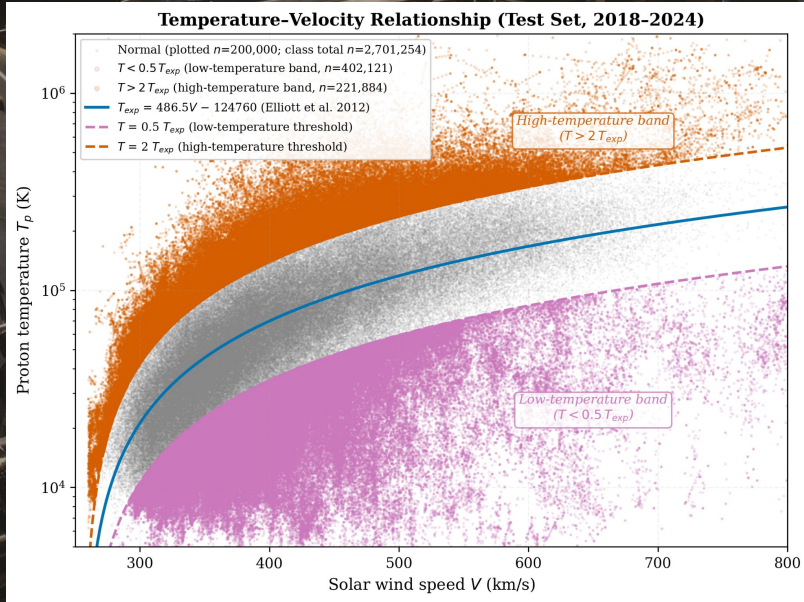
Step	Method
Missing Data	Forward-fill ≤ 2 min; interpolate ≤ 15 min; NaN: longer gaps
Transform	$\log(n_p, V, T_p)$ and z-score standardization
Windowing	7×60 -min windows; 15-min step for training; 1-min step for inference

Physical Scoring Criteria

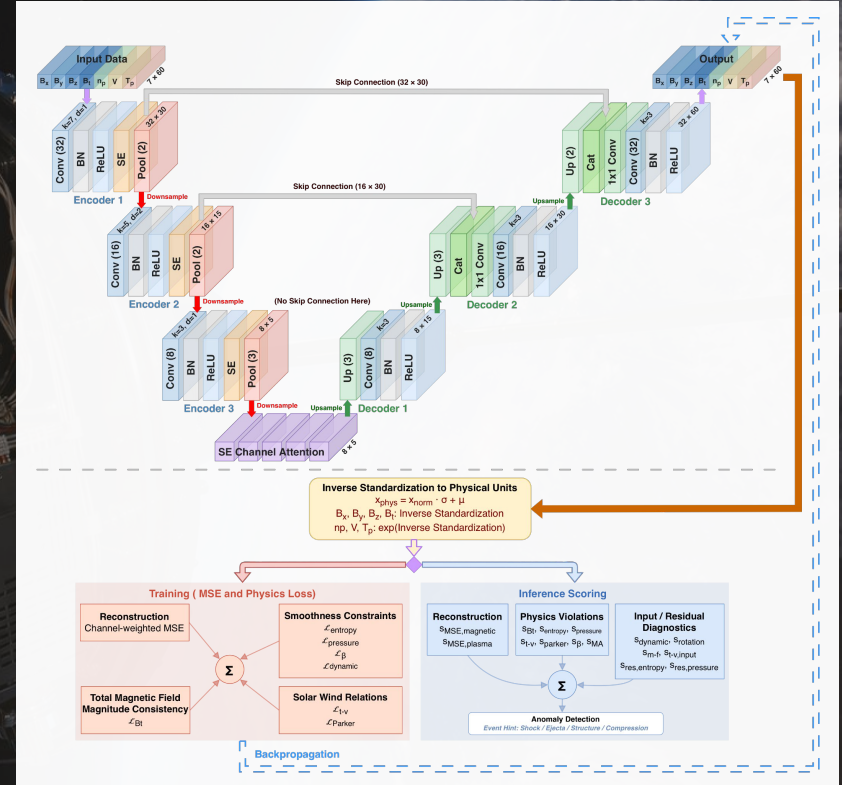
Class	Signatures
Quiet Solar Wind	Smooth entropy, thermal pressure, magnetic pressure, plasma β
Shock/Compression	Abrupt jumps in total pressure, dynamic pressure, and mass flux
ICME	$\beta < 0.5$; magnetic field rotation; Parker-spiral and T–V deviation

Data and Training Summary

Methods and Model



Example Constraint: T-V Relationship

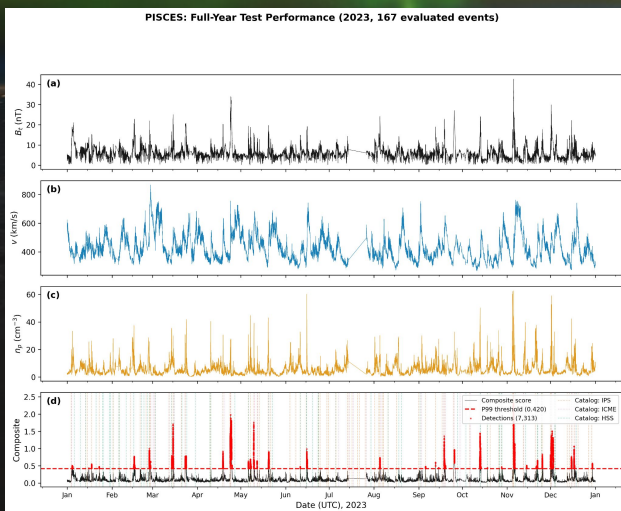


PISCES Architecture and Anomaly Detection Pipeline

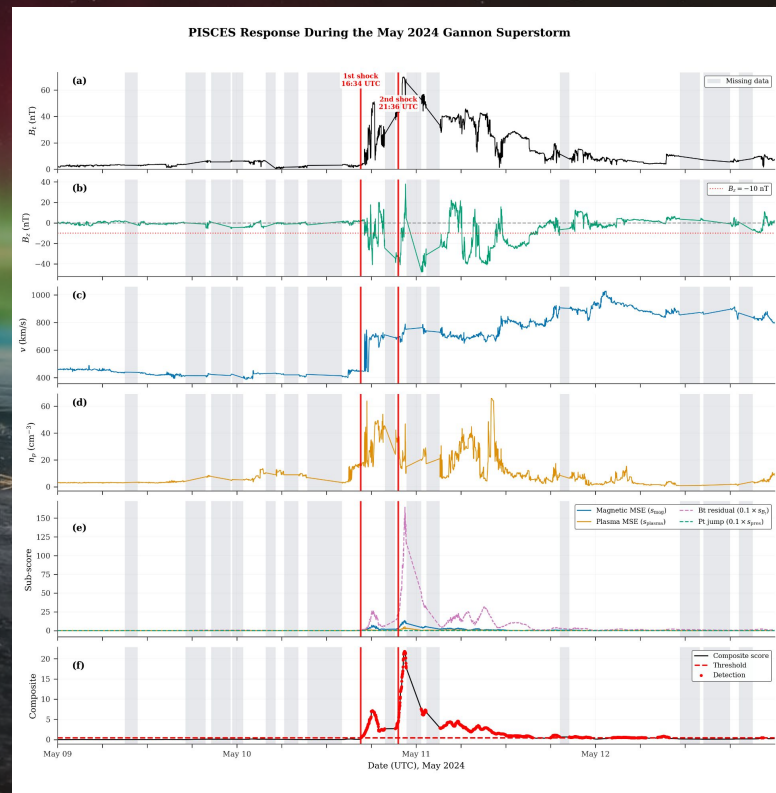
Results and Case Study

Method	PR-AUC	POD	FAR	Precision	HSS
PISCES	0.349	0.059	0.305	0.695	0.092
OC-SVM	0.256	0.082	0.328	0.672	0.124
PCA	0.228	0.039	0.437	0.563	0.058
Conv-VAE	0.222	0.035	0.440	0.560	0.053
LSTM-AE	0.214	0.038	0.445	0.555	0.056
Max-Jump	0.190	0.054	0.614	0.386	0.068

Benchmark of Models and Methodologies



Test Set Performance (2023)



PISCES Response During the May 2024 Gannon Superstorm

Contacts

GitHub Repository*: <http://github.com/magnaprolog/PISCES>

Contacts:

Alison March: alison.march@colorado.edu

Kevin Lee: kevinlee69720@g.ucla.edu



Repo, Model, Q&A

**Code Available after mid-May 2026.*